## 14. How to run Nmap scan

**Step 1: Install Nmap**

Open the terminal.

Run the command:

**sudo apt update**

**sudo apt install nmap**

**Step 2: Identify the Target**

Decide the IP address or hostname you want to scan (e.g., 192.168.1.5).

**Step 3:Run a Basic Nmap Scan**

Type the command in the terminal:

**nmap 192.168.1.5**

This will display the open ports and their associated services on the target.

**Step 4: Perform a Service and Version Scan (Optional)**

For more detail about the running services and their versions, use:

**nmap -sV 192.168.1.5**

This helps in identifying specific software running on open ports.

**Step 5: Interpret the Results**

The terminal will show a list of scanned ports, their status (open/closed/filtered), and common services like SSH (port 22), HTTP (port 80), etc.

Example (sample output):

text

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

443/tcp open  https

